

**BAULKHAM HILLS
HIGH SCHOOL**

2018

**YEAR 11
YEARLY
ASSESSMENT**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board- approved calculators may be used
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work
- A reference sheet is attached at the back of this question paper

**Total marks:
70**

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6 – 10)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Section I

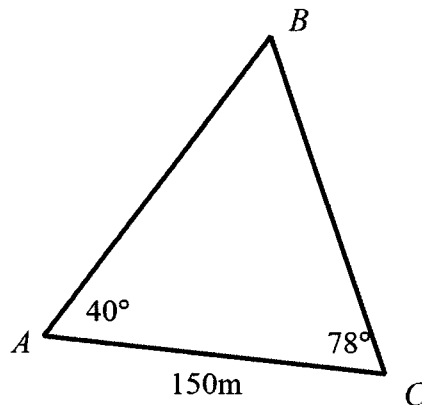
10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

- 1 The length of BC correct to the nearest metre is:

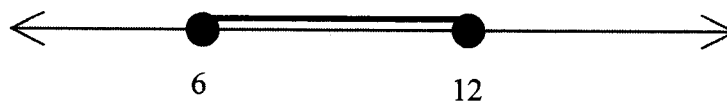


- (A) 99 m
- (B) 109 m
- (C) 166 m
- (D) 245 m

- 2 What is the domain and range of the function $f(x) = \sqrt{4 - x^2}$?

- | | | | | |
|-----|---------|--------------------|--------|--------------------|
| (A) | Domain: | $-2 \leq x \leq 2$ | Range: | $-2 \leq y \leq 2$ |
| (B) | Domain: | $-2 \leq x \leq 2$ | Range: | $0 \leq y \leq 2$ |
| (C) | Domain: | $0 \leq x \leq 2$ | Range: | $-2 \leq y \leq 2$ |
| (D) | Domain: | $0 \leq x \leq 2$ | Range: | $0 \leq y \leq 4$ |

- 3 The number line represents the solution to the inequality $|x - a| \leq b$.



What are the values of a and b ?

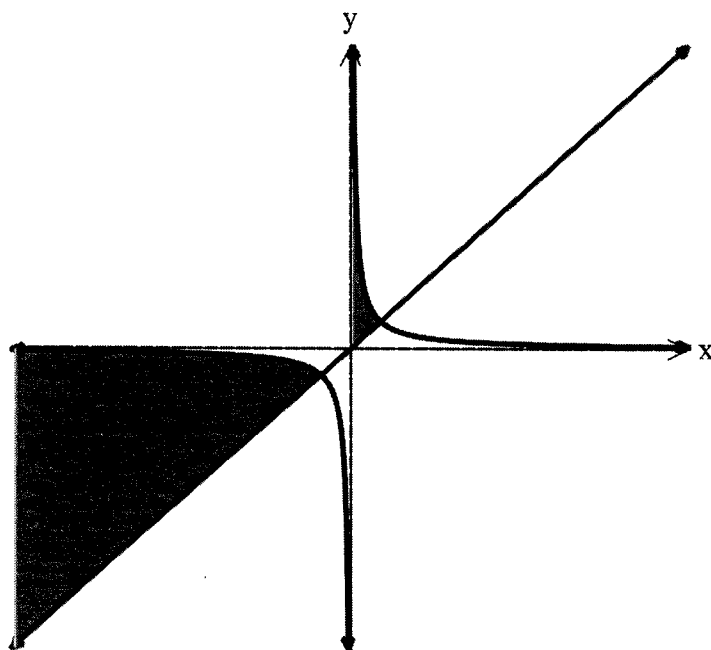
- (A) $a = 6, b = 3$
- (B) $a = 6, b = 6$
- (C) $a = 9, b = 3$
- (D) $a = 9, b = 6$

- 4 What is the value of $f'(2)$ if $f(x) = \frac{1}{3x}$?
- (A) $-\frac{3}{4}$
- (B) $-\frac{1}{6}$
- (C) $-\frac{1}{12}$
- (D) $\frac{1}{3}$
- 5 A function $y = f(x)$ has $f'(3) = 0$ and $f''(3) = -1$. At the point where $x = 3$, $y = f(x)$ is:
- (A) Stationary and concave up
- (B) Decreasing and concave up
- (C) Stationary and concave down
- (D) Stationary with a horizontal point of inflexion.
- 6 How many values of x satisfy the equation $(\sin x + 1)(\tan^2 x - 3) = 0$ for $0^\circ \leq x \leq 180^\circ$?
- (A) 1
- (B) 2
- (C) 3
- (D) 4

7 Let $f(x) = x^4 - 1$. Which of the following statements is NOT true?

- (A) $f(x)$ is an even function.
- (B) $f''(x) \geq 0$ for all values of x
- (C) $f(x)$ has a horizontal point of inflexion when $x = 0$
- (D) $\lim_{x \rightarrow \infty} \frac{1}{f(x)} = 0$

8 Which of the following inequalities could describe the shaded region below?

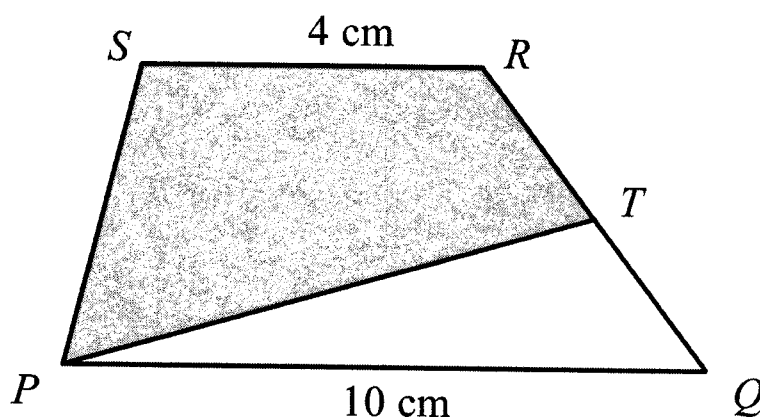


- (A) $y \leq \frac{1}{x}$ and $y \leq x$
- (B) $y \leq \frac{1}{x}$ and $y \geq x$
- (C) $y \geq \frac{1}{x}$ and $y \leq x$
- (D) $y \geq \frac{1}{x}$ and $y \geq x$

- 9 The quadratic equation $3x^2 - 5x + 2 = 0$ has roots α and β . Which of the following statements is true?

- (A) $2\alpha\beta = -\frac{4}{3}$
(B) $\alpha^2 + \beta^2 = \frac{13}{9}$
(C) $2\alpha + 3\beta = \frac{25}{3}$
(D) $\alpha^2\beta^2 = \frac{2}{9}$

- 10 The lines PQ and SR are parallel and 6 cm apart. T is the midpoint of QR .



What is the area, in square centimetres, of the shaded region?

- (A) 21
(B) 26
(C) 27
(D) 34

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour 45 minutes for this section

Answer each question on the appropriate pages of the answer booklet. Extra writing paper is available.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start on the appropriate page of your answer booklet.

- a) Evaluate $\left(\frac{\sqrt{17}}{6 \times 10^{-2}}\right)^3$, correct to three significant figures. 2
- b) Factorise $a^2 - b^2 + 2a + 2b$ 2
- c) Solve the inequality $|2x - 1| > 3$. 2
- d) Simplify $\frac{5}{m-2} - \frac{2}{m-3}$. 2
- e) Solve the following simultaneous equations: 2
 $3x + 6y = 7$
 $2x + 9y = 3$
- f) Evaluate $\lim_{x \rightarrow 3} \frac{3 + 2x - x^2}{x - 3}$ 2
- g) Solve $|3 - x| = x + 1$. 3

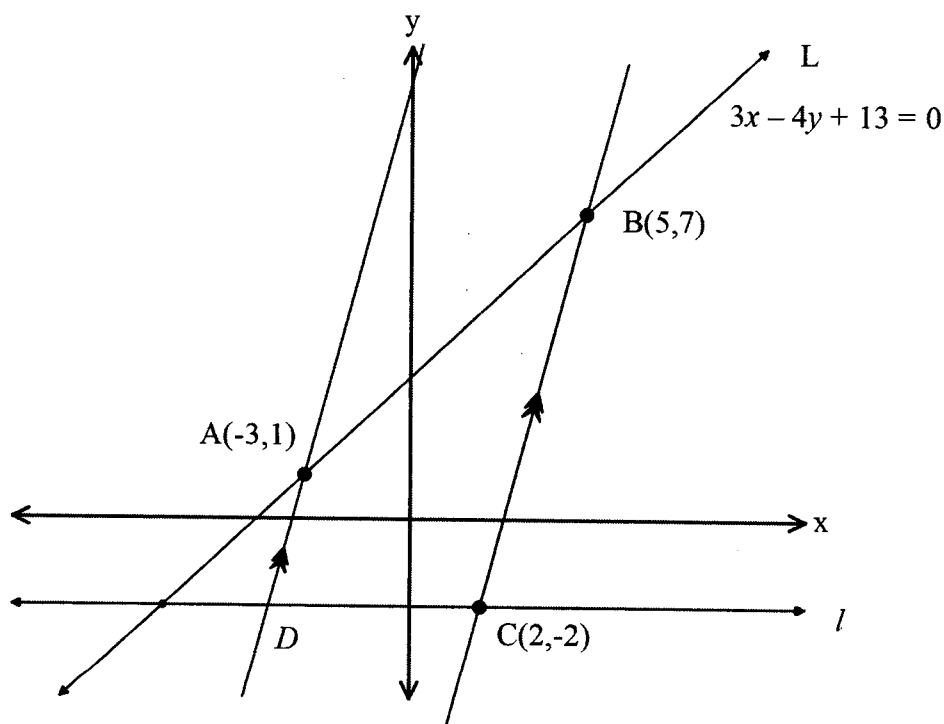
End of Question 11

Question 12 (15 marks) Start on the appropriate page of your answer booklet.

a) Express the following with a rational denominator $\frac{4\sqrt{2}-3}{5+\sqrt{2}}$. 2

b) If $\sec \theta = \frac{5}{4}$, and θ is acute, find the value of $\cot(90^\circ - \theta)$. 2

- c) The points $A(-3,1)$ and $B(5,7)$ lie on the line L with equation $3x - 4y + 13 = 0$.
The line l is parallel to the x -axis. The points $C(2,-2)$ and D are two points on l such that $DA \parallel CB$.

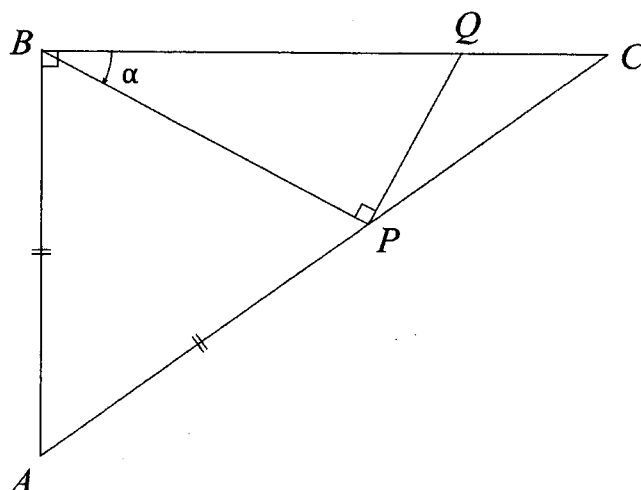


- | | |
|---|---|
| i) Calculate the distance AB . | 1 |
| ii) Find the perpendicular distance of C to the line L . | 2 |
| iii) Show that the equation of the line AD is $y = 3x + 10$. | 2 |
| iv) Find the coordinates of D . | 1 |
| v) By joining AC , find the area of quadrilateral $ABCD$. | 2 |
- d) Solve $2\sin^2 \theta = \sin \theta$ for $0^\circ \leq \theta \leq 360^\circ$. 3

End of Question 12

Question 13 (15 marks) Start on the appropriate page of your answer booklet.

- a) Consider the function $f(x) = 1 + 3x - x^3$ for $-2 \leq x \leq 2$.
- Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. 3
 - Find the coordinates of any points of inflection. 2
 - Sketch the curve for $-2 \leq x \leq 2$, clearly labelling any stationary points and the y -intercept. 2
- b) For the parabola $(x + 2)^2 = 12(y - 6)$, find:
- The coordinates of the focus. 2
 - The equation of the directrix. 1
- c) Triangles ABC and BPQ are right-angled triangles and $AB = AP$. $\angle PBQ = \alpha^\circ$

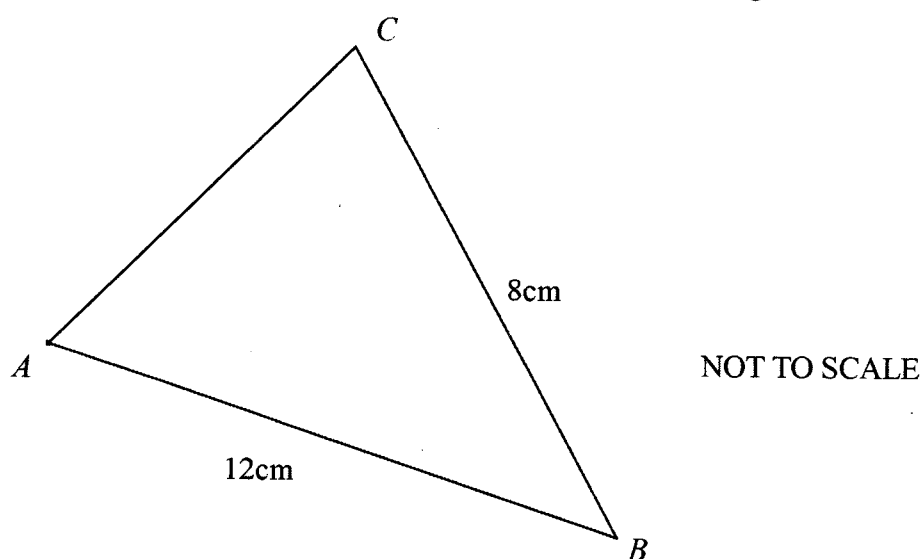


- Copy the diagram into your answer booklet.
- Prove that $\angle PBQ = \angle CPQ$. 2
- Hence prove that $PC^2 = BC \times QC$. 3

End of Question 13

Question 14 (15 marks) Start on the appropriate page of your answer booklet.

- a) Find the value(s) of a such that one of the roots of $x^2 + 4ax + 18a = 0$ is equal to three times the other. **2**
- b) Show that the curve $y = \frac{3x + 5}{x + 1}$ is always decreasing for all values of x in its domain. **3**
- c) The quadratic expression $(3 - k)x^2 + kx + 1$ is positive definite. Find the possible values of k . **3**
- d) In a triangle ABC the length of side AB is 12 cm and the length of side BC is 8 cm.



Within what range of values does the length of side AC lie? **2**

Question 14 continues on following page

Question 14 (continued)

- e) A truck is to travel 1000 kilometres at a constant speed of v km/h. When travelling at v km/h, the truck consumes fuel at the rate of $\left(6 + \frac{v^2}{50}\right)$ litres per hour.

The truck company pays \$1.50 per litre for fuel and pays each of 2 drivers \$36 per hour whilst the truck is travelling.

- i) Let the total cost of fuel and the drivers' wages for the trip be C dollars.

Show that $C = 30v + \frac{81000}{v}$. 2

- ii) The truck must take no longer than 12 hours to complete the trip, and speed limits require that $v \leq 100$.

At what speed v should the truck travel to minimise the cost C ? 3
(Answer to 2 decimal places).

End of paper

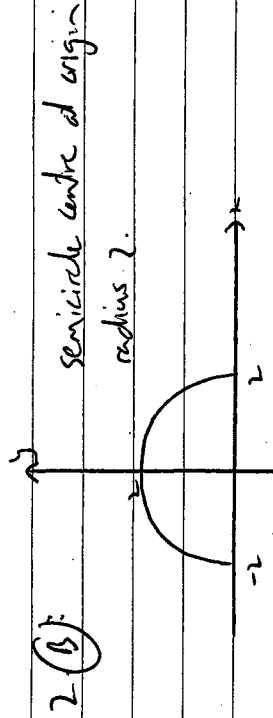
1. (B) $\angle ABC = 180 - (40 + 78)$
 $= 62^\circ$

$BC = \frac{AC}{\sin 40^\circ} \sin 62^\circ$

$BC = \frac{150 \sin 40^\circ}{\sin 62^\circ}$

$\approx 109.200 \dots$

$BC = 109m$ (nearest metre)



3. (C): $|x-9| \leq 3 \Rightarrow x \in [6, 12]$, centre at 9

4. (C) $f(x) = \frac{1}{3}x^{-1}$

$f'(x) = -\frac{1}{3}x^{-2}$

$f''(x) = \frac{2}{3x^3}$

$f'(2) = -\frac{1}{12}$

$\approx -\frac{1}{12}$

5. (C) $f'(3) = 0 \Rightarrow$ stationary point
 $f''(3) = -1 < 0 \Rightarrow$ concave down
 $\therefore$ maximum turning point

6. (B): $(\sin x + 1)(\tan^2 x - 3) = 0$
 $\sin x = -1$ $\tan^2 x = 3$

no solutions since $\sin x > 0$ for $0^\circ < x < 180^\circ$
 $\tan x = \pm\sqrt{3}$
 $\therefore x = 60^\circ, 120^\circ$

$\therefore$ 2 solutions

7. (C): Consider (A) $f(-a) = (-a)^4 - 1$
 $= a^4 - 1$

$= f(a)$

$\therefore$ even $\Rightarrow$ (A) is true

Consider (B) $f'(x) = 4x^3$
 $f''(x) = 12x^2$

$\therefore f''(x) \geq 0$ for all x

Consider (C) $f''(x) = 0$
 $12x^2 = 0$
 $x = 0$

Possible point of inflexion when $x=0$

x	-1	0	1
$f''(x)$	12	0	12

concavity doesn't change

$\therefore$ no point of inflexion

$\therefore$ (C) is false

(Note: Consider (D) $\lim_{x \rightarrow \infty} \frac{1}{f(x)}$
 $= \lim_{x \rightarrow \infty} \left(\frac{1}{x^4 - 1} \right)$
 $= 0$

$\therefore$ D is true

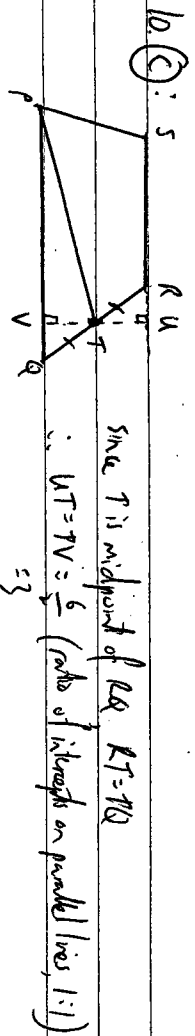
8 (B): Region lies above $y=2$ ie $y \geq 2$
 Region lies below $y=\frac{1}{x}$ ie $y \leq \frac{1}{x}$

9 (B) $\int_0^1 (1-x) dx = \frac{1}{2}$
 $2 \times \frac{1}{2} = \frac{1}{2}$ is not true

For (B) $x^2 + y^2 = (x+1)^2 - 2x+1$
 $= \left(\frac{x}{2}\right)^2 - 2 \times \frac{x}{2}$
 $= \frac{1}{4}$ is true

For (C) $2x+3y = 2(x+1y) + 1y$
 $\therefore 2x+3y = \frac{25}{3}$ if true
 $p = \frac{25}{3}$
 And $3x+5y = 5x+12 = 5x+12$
 $\neq 0$

$\therefore p = 5$ is not a root.
 $\therefore 2x+3y \neq \frac{25}{3}$ is not true
 For (D) $x^2 + y^2 = (x+y)^2$
 $= \left(\frac{x}{2}\right)^2$
 $= \frac{1}{4}$ is not true.



Area of $\triangle PQR = \frac{1}{2} (4 \times 10) = \frac{1}{2} \times 40 = 20$
 $= 42 - 15$
 $= 27 \text{ cm}^2$

11 (a) 324503.6815
 ≈ 325000 significant figures
 or 3.25×10^5
 (B) correct answer
 (D) obtains correct unrounded answer.

(b) $a^2 - b^2 + 12a + 12b$
 $= (a+b)(a-b) + 12(a+b)$
 $= (a+b)(a-b+12)$
 (C) correct answer
 (D) function is $a^2 - b^2$

(c) $2x-1 < -3$ or $2x-1 > 3$
 $2x < -2$ or $2x > 4$
 $x < -1$ or $x > 2$
 (D) correct answer
 (E) solves inequality correctly

(d) $\frac{5(m-3) - 2(m-2)}{(m-2)(m-3)}$
 $= \frac{5m-15-2m+4}{(m-2)(m-3)}$
 $= \frac{3m-11}{(m-2)(m-3)}$
 (B) correct answer
 (D) identifies common denominator

(e) $\begin{cases} x+1y=7 \\ 2x+9y=3 \end{cases}$ (1) correct solution
 (2) finds one correct value

(f) $\begin{cases} 6x+12y=14 \\ 6x+12y=9 \end{cases}$ (3)
 (4) $15y = -5$
 $y = -\frac{1}{3}$
 sub in (1) $3x-2=7$
 $3x=9$
 $x=3$

$\therefore x=3$ and $y=-\frac{1}{3}$

11 (b) $\lim_{n \rightarrow 3} \frac{(3-n)(1+n)}{n-3}$
 $= \lim_{n \rightarrow 3} \frac{(3-n)(1+n)}{n-3}$
 $= \lim_{n \rightarrow 3} \frac{-(1+n)}{1}$
 $= -(1+3)$
 $= -4$

② correct solution
 ① factorises numerator

(9) $|3-x| = x+1$

note $x+1 \geq 0$ i.e. $x \geq -1$

$3-x = x+1 \quad - (3-x) = x+1$

$2x = 2 \quad x-3 = x+1$

$x = 1 \quad -3 \neq 1$

$\therefore$ no solution

test/check since $x=1$ satisfies $x \geq -1$ it is a solution

$\therefore x=1$ is the only solution

③ correct solution

② defines $x=1$ and no solution without testing

① solves for $x=1$

12 (a) $\frac{4\sqrt{2}-3}{5\sqrt{2}} \times \frac{5-\sqrt{2}}{5-\sqrt{2}}$

$= \frac{20\sqrt{2}-8-15\sqrt{2}}{25-2}$

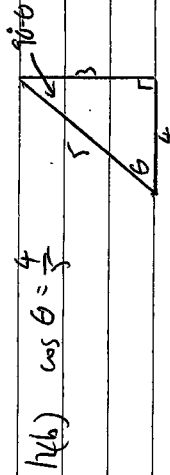
$= \frac{23\sqrt{2}-23}{23}$

$= \frac{23(\sqrt{2}-1)}{23}$

$= \sqrt{2}-1$

② correct answer $\frac{23\sqrt{2}-23}{23}$ is acceptable

① multiplies by conjugate



② correct answer

① finds $\tan \theta$

or ① attempts to use complementary relationship between $\sin$ and $\tan$

$\cos(90-\theta) = \tan \theta$
 $= \frac{3}{4}$

12 (c) (i) $AD = \sqrt{(5-3)^2 + (7-1)^2}$
 $= \sqrt{64+36}$
 $= 10$

① correct answer

(ii) Perp Dist = $\frac{|2 \times 3 + 4 \times 2 + 13|}{\sqrt{3^2 + 4^2}}$
 $= \frac{27}{5}$

① correct answer

① correctly substitutes into formula

(iii) $m_{AC} = \frac{7-2}{5-2}$

$= \frac{5}{3}$

$= \frac{5}{3}$

$= \frac{5}{3}$

② correct solution

① calculates gradient of AD (must mention parallel lines)

$m_{AD} = m_{AC}$ (parallel lines)

$\therefore m_{AD} = \frac{5}{3}$

Using $y-y_1 = m(x-x_1)$

$y-1 = \frac{5}{3}(x-3)$

$y-1 = \frac{5}{3}x-5$

$y = \frac{5}{3}x-4$

(iv) At D, $y = -2$

$-2 = \frac{5}{3}x-4$

$3x = -6$

$x = -2$

$\therefore D$ is $(-2, -2)$

① correct answer

(v) Area $\triangle DAC + \text{Area } \triangle ABC = \text{Area of } ABCD$ ② correct answer

Area = $(\frac{1}{2} \times 6 \times 3) + (\frac{1}{2} \times 10 \times \frac{27}{5})$ ① calculator area of

ADAC or ABC

① holds answer or another

$= 9 + 27$

$= 36$ square units correct method and answer

12 (b) $2\sin\theta - \sin\theta = 0$

$\sin\theta(2\sin\theta - 1) = 0$

$\sin\theta = 0$ $\sin\theta = \frac{1}{2}$

$\theta = 0, 180^\circ, 360^\circ$ $\theta = 30^\circ, 180^\circ - 30^\circ$

$\therefore \theta = 0^\circ, 30^\circ, 150^\circ, 180^\circ, 360^\circ$ and find at least 1 solution

(3) correct answer

(2) solves sine $\theta = \frac{1}{2}$ to obtain $30^\circ, 150^\circ$

(1) obtains $\sin\theta = 0$ and $\sin\theta = \frac{1}{2}$

Question 13

Q13 (a) $f(x) = 13x - x^3$

$f'(x) = 13 - 3x^2$

$f''(x) = -6x$

Set points occur when $\frac{dy}{dx} = 0$

$13 - 3x^2 = 0$

$3x^2 = 13$

$x^2 = \frac{13}{3}$

$x = \pm \sqrt{\frac{13}{3}}$

Testing nature:

$f''(x) = 6 > 0 \quad \cup$

$\therefore$ Local minimum turning point at $(-\sqrt{\frac{13}{3}}, -13\sqrt{\frac{13}{3}})$

$f''(x) = -6 < 0 \quad \cap$

$\therefore$ Local maximum turning point at $(\sqrt{\frac{13}{3}}, 13\sqrt{\frac{13}{3}})$

(ii) Point of inflexion may occur when $\frac{d^2y}{dx^2} = 0$

$-6x = 0$

$x = 0$

Testing

x	-1	0	1
$f''(x)$	6	0	-6

Concavity changes $\therefore$ Point of inflexion at $(0, 13)$

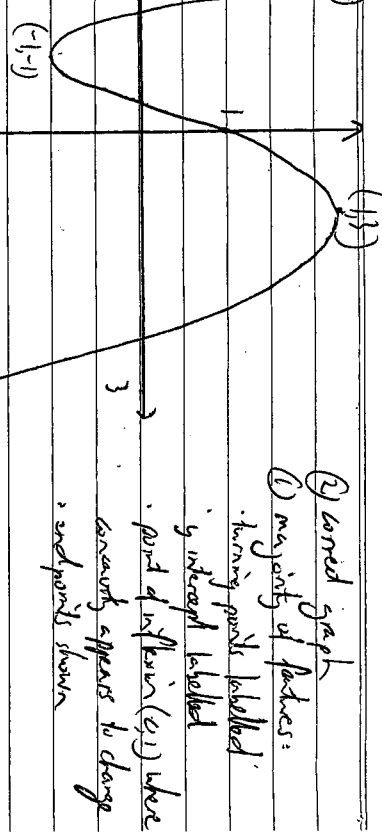
(3) correct solution

(1) finds coordinates without showing concavity changes

13 (a) (iii)

$(-1, 3)$

$(1, 1)$



(2) correct graph

(1) majority of features:

turning points labelled

y-intercept labelled

point of inflection $(1, 1)$ where concavity appears to change

end points shown

$f(-2) = 3$

$f(3) = -17$

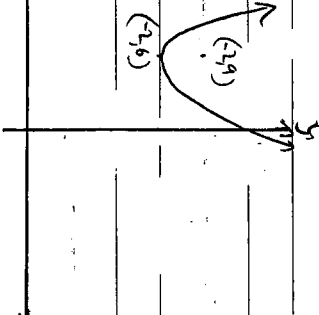
b (i) $4a = 12$

$a = 3$

vertex at $(2, 6)$

$(-1, 9)$

$(3, 6)$



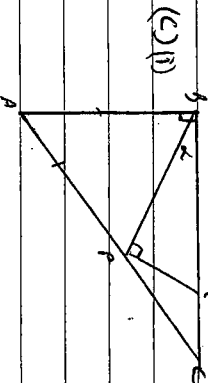
(2) correct answer

(1) calculates focal length (must be positive a)

$\therefore$ focus at $(2, 9)$

(ii) $y = 3$

(1) correct answer



(ii) $\angle CBP + \angle PBA = 90^\circ$ (complementary adjacent angles)

$\angle PBA = 90^\circ - \alpha$

$\angle BPA = 90^\circ - \alpha$

$\angle APB + \angle BPA + \angle APC = 180^\circ$ ($\angle$ sum on straight line)

$90^\circ - \alpha + 90^\circ + \angle APC = 180^\circ$

$\therefore \angle APC = \alpha = \angle QBP$ as required

(2) correct solution

(1) significant prog

13c(iii) In $\triangle ABC$ $\angle A < 90^\circ$ ① correct proof
 $\angle C < 90^\circ$ (given in (ii)) ② correct logic without reasons
 $\angle C$ is common ③ proves $\triangle ABC \parallel \triangle DEC$ with reasons
 $\therefore \triangle ABC \parallel \triangle DEC$ (AA) ④ limited progress (eg incomplete)
 $\therefore BC \parallel DE$ (making sides in same ratio, $\triangle ABC \parallel \triangle DEC$ similarly proved)
 $\frac{BC}{DE} = \frac{AC}{EC}$
 $\therefore PC^2 = BC \cdot AC$

Q14(a) let roots be α and 3α ② correct answer (no penalty if finding $\alpha=0$ and then ignoring as trivial)
 $\alpha + 3\alpha = -4a$
 $4\alpha = -4a$
 $\alpha = -a$
 Also $\alpha(3\alpha) = 18a$
 $3\alpha^2 = 18a$
 $\alpha^2 = 6a$
 $a^2 = 6a$
 $a(a-6) = 0$
 $\therefore a = 0$ or $a = 6$

(b) $\frac{dy}{dx} = \frac{(x+1) \cdot 3 - (3x+5)}{(x+1)^2}$ ③ correct solution
 $= \frac{3x+3-3x-5}{(x+1)^2}$ ④ finds simplified $\frac{dy}{dx}$
 $= \frac{-2}{(x+1)^2}$ ⑤ correctly applies the quotient rule
 $\frac{dy}{dx} = \frac{-2}{(x+1)^2}$

Always decreasing as negative $\div$ by positive $(x+1)^2 > 0$ is always negative except $x = -1$ where the denominator is 0

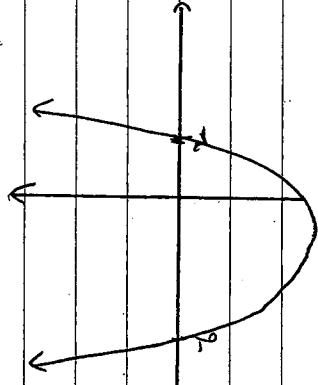
14(c) For positive definite

$a > 0$ and $\Delta < 0$

$3-k > 0$ and $k^2 - 4(3-k) \cdot 1 < 0$

$k < 3$ $k^2 + 4k - 12 < 0$

$(k+6)(k-2) < 0$



$k < 3$ and $-6 < k < 2$

$\therefore -6 < k < 2$

14(d) Since the sum of any 2 sides of a triangle must exceed the third side

$AC + 18 > 12$ and $12 + 18 > AC$

$AC > 4$ $AC < 20$

$\therefore 4 < AC < 20$ cm

① either $AC > 4$ or $AC < 20$

② correct answer

14 a) i) $\text{speed} = \frac{\text{distance}}{\text{time}}$

$\text{time} = \frac{v}{1000}$

$\text{cost per hour} = \text{driver cost/hr} + \text{fuel cost/hr}$
 $= 2 \times 36 + \left(6 + \frac{v^2}{50}\right) \times 1.5$

$= 81 + 3 \frac{v^2}{100}$

$\text{Total cost} = \left(81 + 3 \frac{v^2}{100}\right) \times \frac{v}{1000}$

$C = 30v + \frac{3v^3}{1000}$

ii) $\frac{dC}{dv} = 30 - \frac{9v^2}{1000} = 0$

$30 = \frac{9v^2}{1000}$

For max or min, $\frac{dC}{dv} = 0$

$30 - \frac{9v^2}{1000} = 0$

$30 = \frac{9v^2}{1000}$

$v^2 = 2700$

$v = 51.96 \text{ km/hr}$

$\frac{d^2C}{dv^2} = \frac{18000}{v^3}$

when $v = 51.96$, $\frac{d^2C}{dv^2} = 1.1547 > 0$

∴ min cost

∴ trip must be less than 12 hours $\frac{1000}{51.96} \approx 19.25 \text{ hr}$

$v \approx 83.3 \text{ km/hr}$

∴ Time constant is not met by minimum cost. As speed increases cost increases so 83.3 km/hr is the speed the truck should travel.

(2) correct solution

(1) calculates cost per hour

(3) correct solution

(2) uses calculus to find

minimum cost at 51.96 km/hr

proving it is a minimum

(1) obtains 51.96 by solving

$\frac{dC}{dv} = 0$

(Note: an answer of 83.3 km/hr)

(without any calculus is 0 marks)